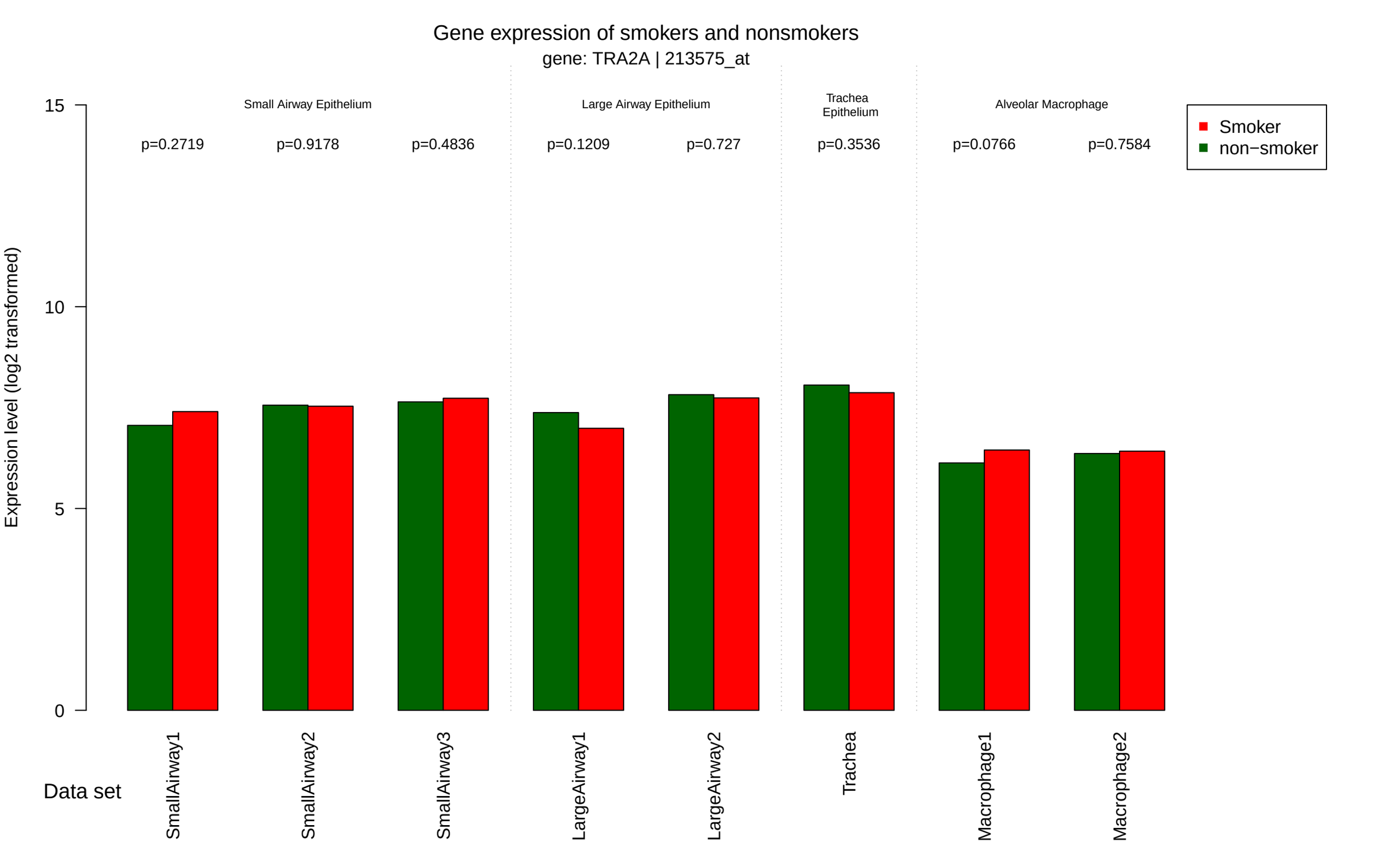
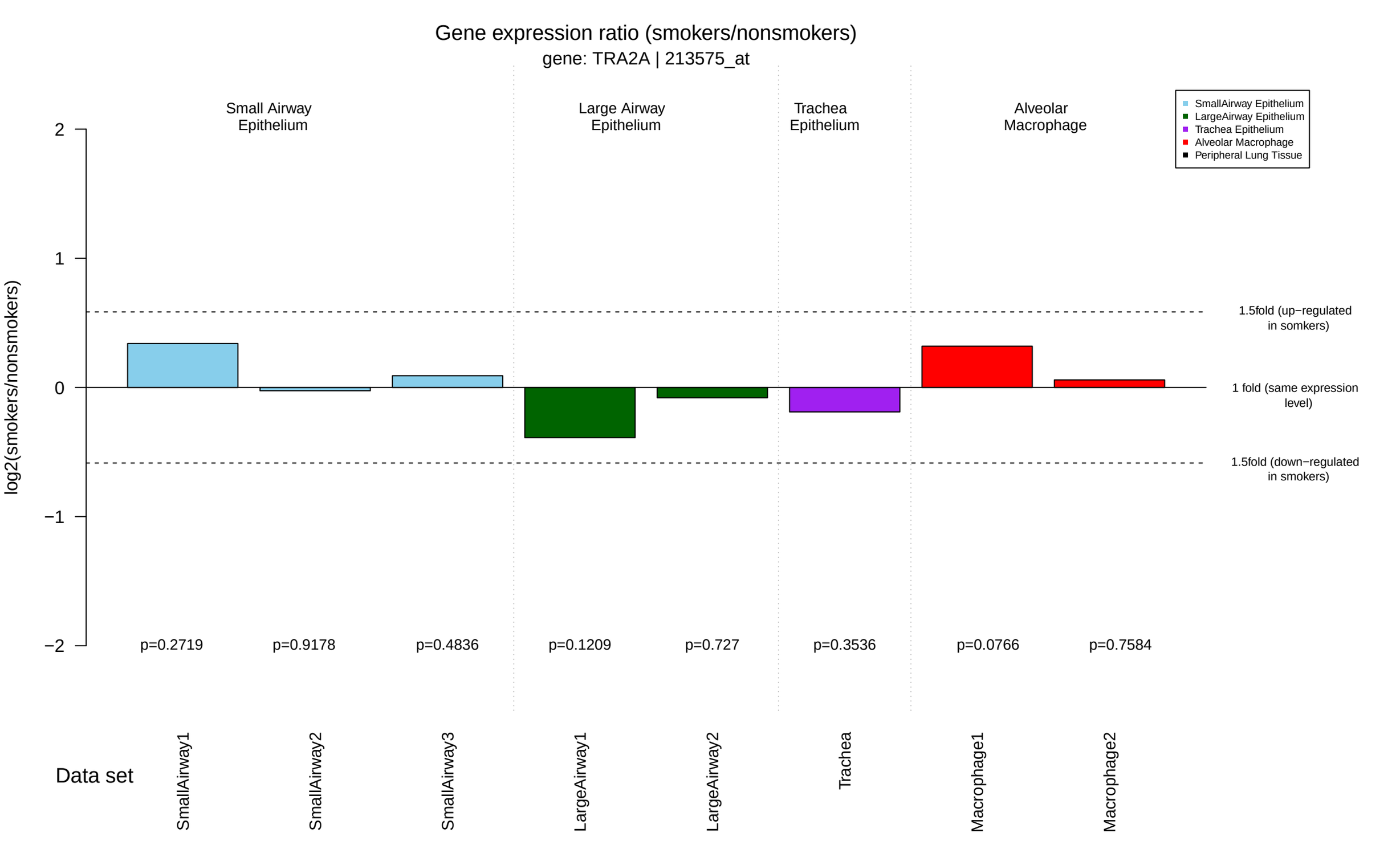
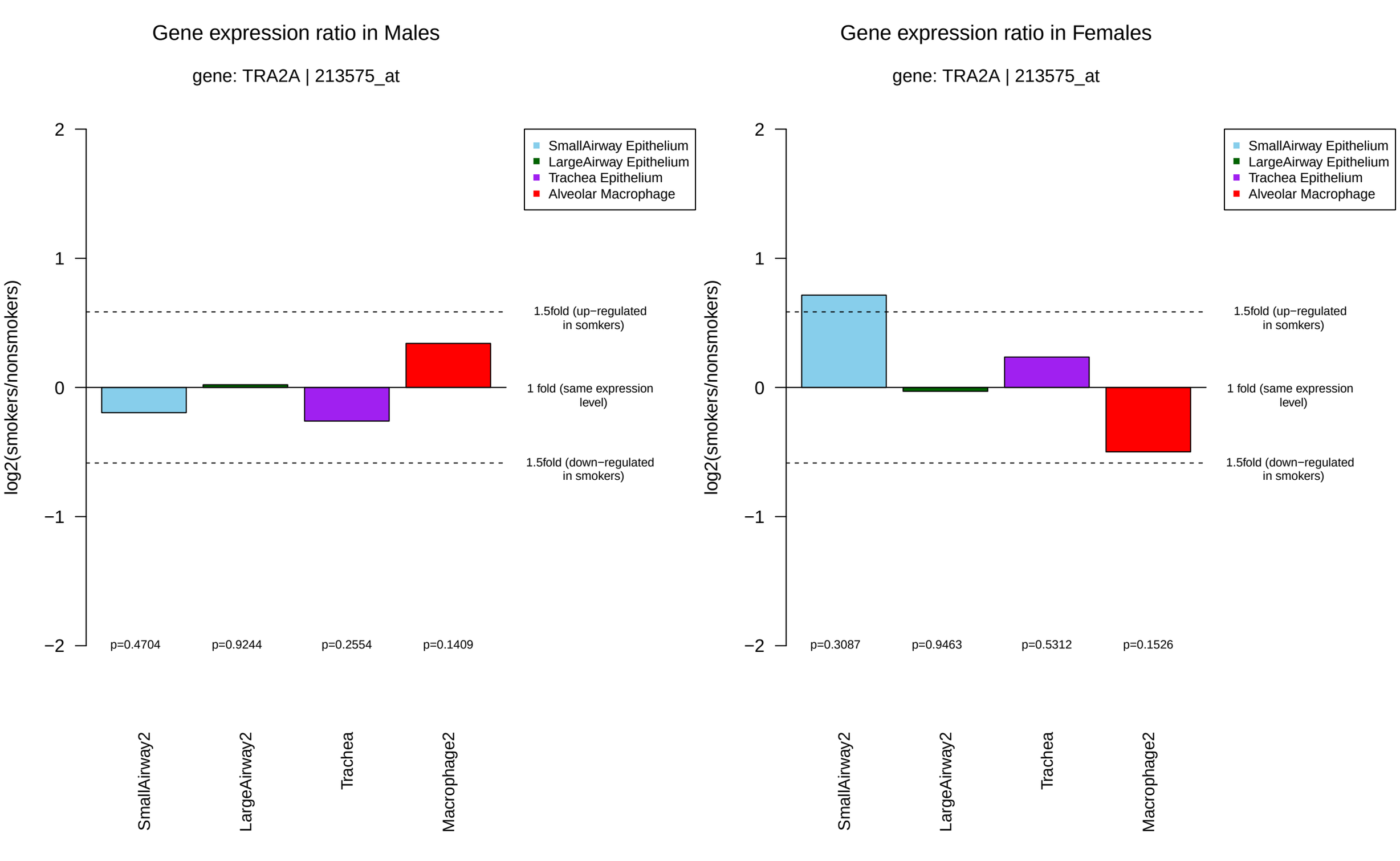
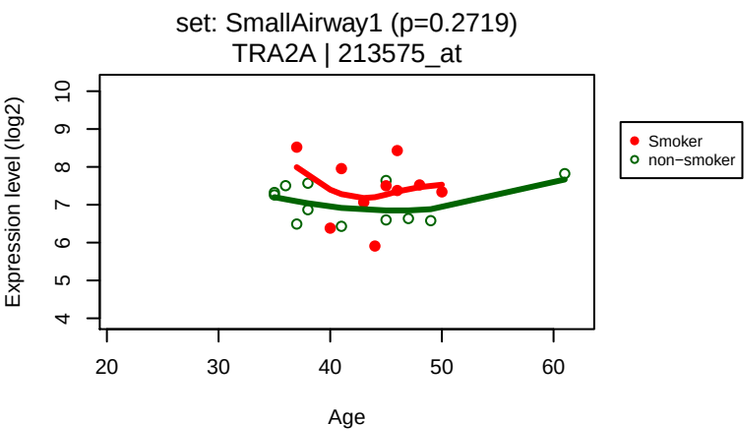
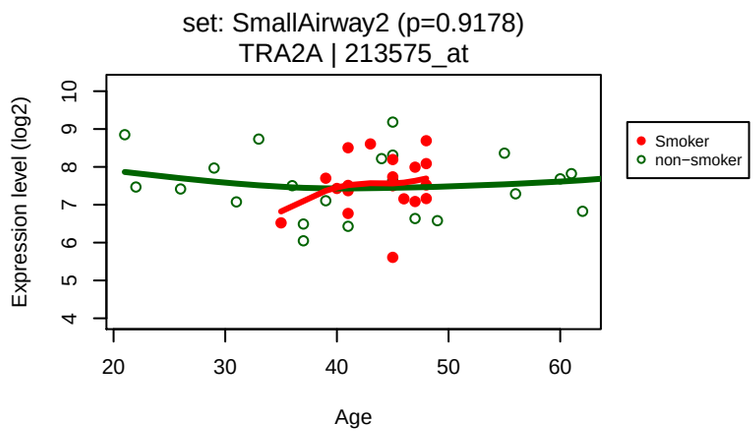
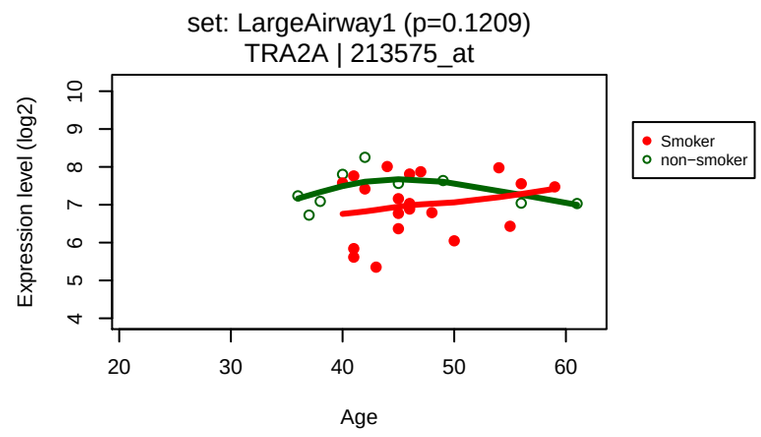

Gene expression of smokers and nonsmokers
gene: TRA2A | 213575_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2719
p=0.9178
p=0.4836
p=0.1209
p=0.727
p=0.3536
p=0.0766
p=0.7584
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

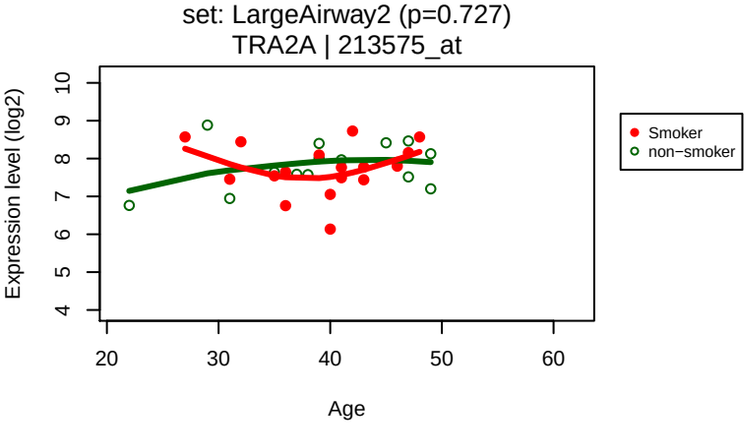

Gene expression ratio (smokers/nonsmokers)
gene: TRA2A | 213575_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2719
p=0.9178
p=0.4836
p=0.1209
p=0.727
p=0.3536
p=0.0766
p=0.7584
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

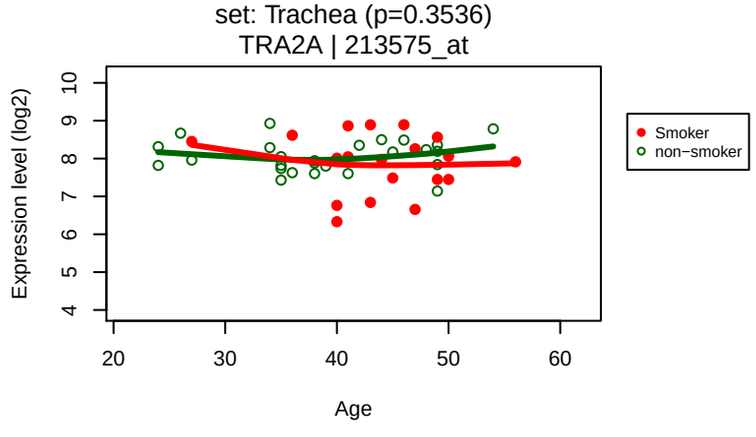

Gene expression ratio in Males
gene: TRA2A | 213575_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4704
p=0.9244
p=0.2554
p=0.1409
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRA2A | 213575_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3087
p=0.9463
p=0.5312
p=0.1526
SmallAirway2
LargeAirway2
Trachea
Macrophage2

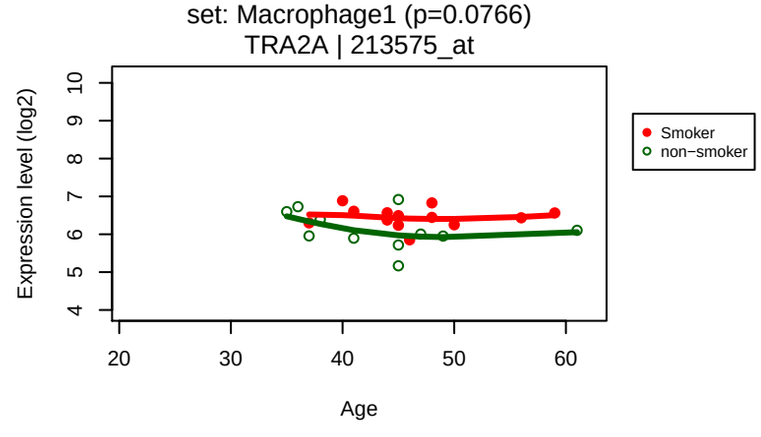

set: SmallAirway1 (p=0.2719)
TRA2A | 213575_at
Expression level (log2)
Age
Smoker
non−smoker

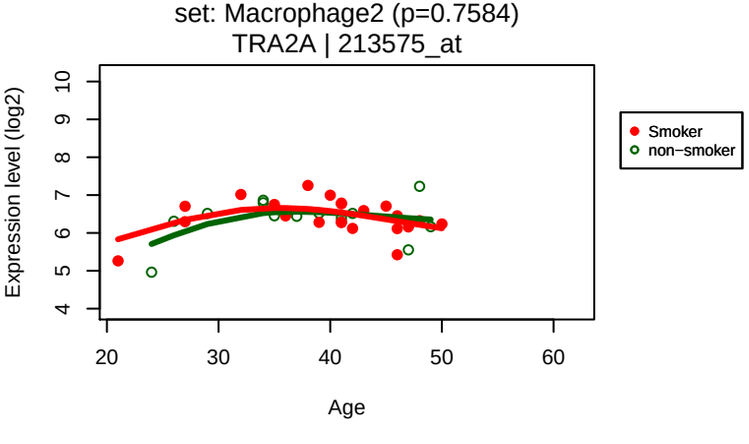

set: SmallAirway2 (p=0.9178)
TRA2A | 213575_at
Expression level (log2)
Age
Smoker
non−smoker

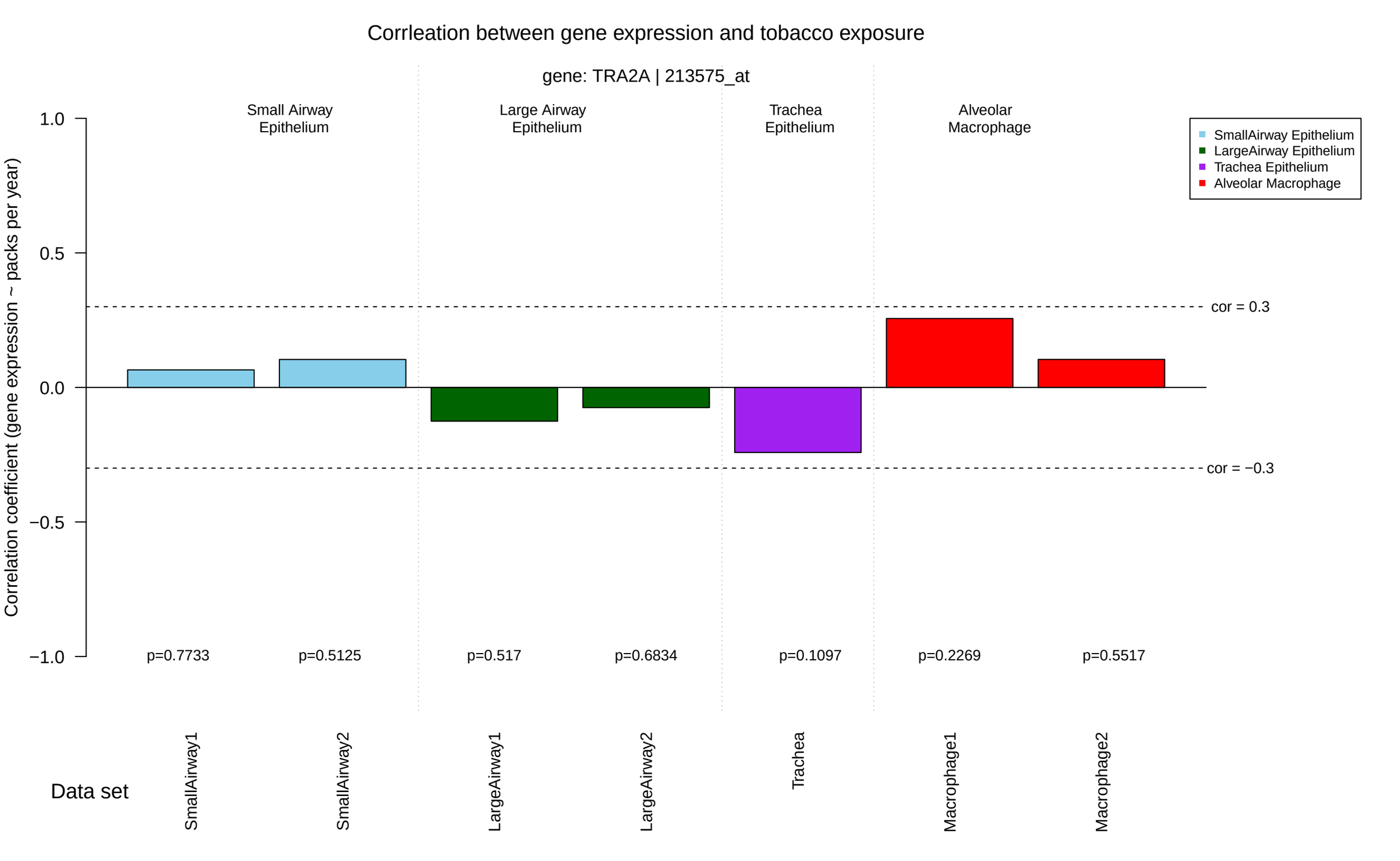

set: LargeAirway1 (p=0.1209)
TRA2A | 213575_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.727)
TRA2A | 213575_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3536)
TRA2A | 213575_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0766)
TRA2A | 213575_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7584)
TRA2A | 213575_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRA2A | 213575_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7733
p=0.5125
p=0.517
p=0.6834
p=0.1097
p=0.2269
p=0.5517
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2